

Security: Deserialization

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Content

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Security: Deserialization

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Deserializing data can trigger the execution of gadgets that may initiate RCE (Remote Code Execution) attacks on the client machine.

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To handle this threat, RavenDB's default deserializer blocks the deserialization of known [.NET RCE gadgets](#).

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Users can customize the list of namespaces and types for which deserialization is forbidden or allowed.

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When a RavenDB client uses the [Newtonsoft library](#) to deserialize a JSON string to a .NET object, the object may include a reference to a **gadget** (a code segment) and the deserialization process may execute it.

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Some gadgets attempt to exploit the deserialization process and initiate an RCE (Remote Code Execution) attack that may, for example, inject the system with malicious code, steal information, or take control of the machine.

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To prevent such exploitation, RavenDB's default deserializer blocks deserialization for the `Microsoft.VisualStudio` namespace and for the following known .NET RCE gadgets:

- `System.Configuration.Install.AssemblyInstaller`
- `System.Activities.Presentation.WorkflowDesigner`
- `System.Windows.ResourceDictionary`
- `System.Windows.Data.ObjectDataProvider`
- `System.Windows.Forms.BindingSource`
- `Microsoft.Exchange.Management.SystemManager.WinForms.ExchangeSettingsProvider`
- `System.Data.DataViewManager`
- `System.Xml.XmlDocument`
- `System.Xml.XmlDataDocument`
- `System.Management.Automation.PSObject`

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Users can [customize](#) the list of namespaces and types for which deserialization is forbidden or allowed.

Directly-loaded gadgets cannot be blocked using `DefaultRavenSerializationBinder`. When a gadget is loaded directly, its execution during deserialization is permitted regardless of the binder's forbidden list.

For example, the following `Load` call will succeed regardless of the binder configuration:

```
session.Load<object>("Gadget");
```

Indirectly-loaded gadgets can be blocked using `DefaultRavenSerializationBinder`. When a gadget type name is embedded as a value inside a JSON string, it is resolved only at deserialization time. At that point the binder can intercept and block it. For example, in the following payload, the type is not loaded directly but appears as a JSON value and is resolved only during deserialization:

```
{
  "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35",
  "MethodName": "Start",
  "MethodParameters": {
    "$type": "System.Collections.ArrayList, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089",
    "$values": ["cmd", "/c calc.exe"]
  },
  "ObjectInstance": {
    "$type": "System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089"
  }
}
```

Including `System.Windows.Data.ObjectDataProvider` in the forbidden list prevents this payload from being deserialized and executed.

Use `DefaultRavenSerializationBinder` and its methods to block the deserialization of suspicious namespaces and types, or to explicitly allow the deserialization of trusted types. Create a new `DefaultRavenSerializationBinder` instance, call the registration methods to define your overrides, then configure the instance as a serialization convention as shown in the [example](#).

All `Register*` methods must be called **before** the binder processes its first type. Calling any registration method after the binder has been used throws:

```
InvalidOperationException: "Cannot perform this operation, because binder was already
used."
```

Note that `DefaultRavenSerializationBinder.Instance` is a shared static singleton. Always create your own `new DefaultRavenSerializationBinder()` instance; do not call registration methods on the static instance.

Use `RegisterForbiddenNamespace` to prevent the deserialization of any type belonging to the given namespace.

```
public void RegisterForbiddenNamespace(string @namespace)
```

| ParameterTypeDescription | **@namespace** string The namespace from which deserialization will be blocked. |||||

Attempting to deserialize a type whose namespace is forbidden throws:

```
InvalidOperationException: "Cannot resolve type '<FullName>' because the namespace is on a blacklist due to security reasons. Please customize json deserializer in the conventions and override SerializationBinder with your own logic if you want to allow this type."
```

If the forbidden type appears as a generic argument (e.g. `List<System.Xml.XmlDocument>`), a containing exception is thrown, with the original wrapped as its inner exception:

```
InvalidOperationException: "Generic type: <FullName>, contains a generic argument <FullName> that is blocked from serialization"
```

Use `RegisterForbiddenType` to prevent the deserialization of a specific type.

```
public void RegisterForbiddenType(Type type)
```

| ParameterTypeDescription | **type** Type The type whose deserialization will be blocked. || |||||

Attempting to deserialize a forbidden type throws: `InvalidOperationException: "Cannot resolve type '<FullName>' because the type is on a blacklist due to security reasons. Please customize json deserializer in the conventions and override SerializationBinder with your own logic if you want to allow this type."`

If the forbidden type appears as a generic argument (e.g. `List<SuspiciousClass>`), a containing exception is thrown, with the original wrapped as its inner exception:

```
InvalidOperationException: "Generic type: <FullName>, contains a generic argument <FullName> that is blocked from serialization"
```

Use `RegisterSafeType` to explicitly allow the deserialization of a specific type. A type registered as safe bypasses all restrictions, including user-registered forbidden namespaces, user-registered forbidden types, and the built-in hardcoded forbidden-types list.

```
public void RegisterSafeType(Type type)
```

| ParameterTypeDescription | **type** Type The type whose deserialization will be permitted. | |
| | | | |

The following example creates a `DefaultRavenSerializationBinder` instance, registers a forbidden namespace, a forbidden type, and a trusted type, then applies it to the document store:

```
var binder = new DefaultRavenSerializationBinder();

binder.RegisterForbiddenNamespace("SuspiciousNamespace");

binder.RegisterForbiddenType(typeof(SuspiciousClass));

binder.RegisterSafeType(typeof(TrustedClass));

var store = new DocumentStore

{
    Conventions =

    {
        Serialization = new NewtonsoftJsonSerializationConventions
        {
            CustomizeJsonDeserializer = deserializer =>

                deserializer.SerializationBinder = binder

        }
    }
};
```

In this article

